WORKED SOLUTION

STEP I 2008, Question 4

Pure • Convex Functions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q4

The key idea

Both functions are exponentials, $\mathrm{e}^{h(x)}$, and differentiating twice gives $(h'' + (h')^2)\mathrm{e}^h$. The exponential factor is always positive, so convexity comes down to the sign of $h'' + (h')^2$, a purely trigonometric inequality. In (i), that inequality turns out to compare exactly the two curves we're asked to sketch. In (ii), it simplifies to a comparison of $\sin 2x$ with $\sin 2\alpha$.

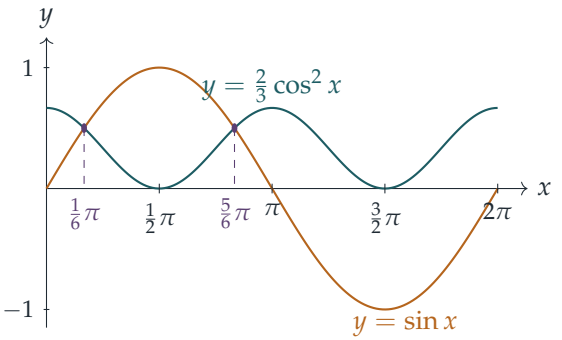
Solution

Part (i)

The sketch. The curve $y = \frac{2}{3}\cos^2 x$ lies between 0 and $\frac{2}{3}$. It equals $\frac{2}{3}$ at $x = 0, \pi, 2\pi$ and touches the axis at $x = \frac{1}{2}\pi, \frac{3}{2}\pi$, with period π. The curve $y = \sin x$ is the usual sine wave. To find where they cross, put $s = \sin x$ and $\cos^2 x = 1 - s^2$:

$$\tfrac{2}{3}(1 - s^2) = s \iff 2s^2 + 3s - 2 = 0 \iff (2s - 1)(s + 2) = 0.$$

Since $s = -2$ is impossible, $\sin x = \frac{1}{2}$, so the curves cross at $x = \frac{1}{6}\pi$ and $x = \frac{5}{6}\pi$ only. On $(\pi, 2\pi)$ the sine curve is negative while the other curve is non-negative, so there are no crossings there.



Convexity of f. With $f(x) = \mathrm{e}^{\frac{2}{3}\sin x}$, the chain rule gives $f'(x) = \frac{2}{3}\cos x\, f(x)$, and then by the product rule

$$\begin{aligned} f''(x) &= -\tfrac{2}{3}\sin x\, f(x) + \tfrac{2}{3}\cos x\, f'(x) \\ &= f(x)\left(\tfrac{4}{9}\cos^2 x - \tfrac{2}{3}\sin x\right) \\ &= \tfrac{2}{3}f(x)\left(\tfrac{2}{3}\cos^2 x - \sin x\right). \end{aligned}$$

Ask yourself Why does the bracket settle everything?

Since $f(x)$ is an exponential, it's strictly positive, so $f''(x) \geq 0$ exactly when $\frac{2}{3}\cos^2 x \geq \sin x$. In other words, f is convex exactly where the teal curve lies on or above the amber one. That's why the question asks for the sketch first.

From the sketch, and the crossing points found above, $\frac{2}{3}\cos^2 x \geq \sin x$ for $0 < x \leq \frac{1}{6}\pi$ and for $\frac{5}{6}\pi \leq x < 2\pi$. Between $\frac{1}{6}\pi$ and $\frac{5}{6}\pi$ the sine curve is above (for instance at $x = \frac{1}{2}\pi$, where the two values are 0 and 1). So f is convex on

$$0 < x \leq \tfrac{1}{6}\pi \qquad \text{and} \qquad \tfrac{5}{6}\pi \leq x < 2\pi,$$

and on no larger interval. (Whether you include the endpoints $\frac{1}{6}\pi$ and $\frac{5}{6}\pi$, where $f'' = 0$, is a matter of taste.)

Part (ii)

With $g(x) = \mathrm{e}^{-k\tan x}$ we have $g'(x) = -k\sec^2 x\, g(x)$. Using $\frac{\mathrm{d}}{\mathrm{d}x}\sec^2 x = 2\sec^2 x\tan x$ and the product rule,

$$\begin{aligned} g''(x) &= -2k\sec^2 x\tan x\, g(x) - k\sec^2 x\, g'(x) \\ &= g(x)\left(k^2\sec^4 x - 2k\sec^2 x\tan x\right) \\ &= k\sec^2 x\, g(x)\left(k\sec^2 x - 2\tan x\right). \end{aligned}$$

Here $k = \sin 2\alpha > 0$ because $0 < 2\alpha < \frac{1}{2}\pi$, and $\sec^2 x > 0$, $g(x) > 0$. So $g''(x) \geq 0$ exactly when $k\sec^2 x \geq 2\tan x$. Multiplying by $\cos^2 x > 0$, this becomes

$$k \geq 2\tan x\cos^2 x = 2\sin x\cos x = \sin 2x.$$

So g is convex exactly where

$$\sin 2x \leq \sin 2\alpha.$$

Ask yourself Where is $\sin 2x \leq \sin 2\alpha$ when $0 < x < \frac{1}{2}\pi$?

As x runs over $(0, \frac{1}{2}\pi)$, $2x$ runs over $(0, \pi)$. On that interval sin increases up to $\frac{1}{2}\pi$ and then decreases, symmetrically about $\frac{1}{2}\pi$. The value $\sin 2\alpha$ (with $0 < 2\alpha < \frac{1}{2}\pi$) is

reached at $2x = 2\alpha$ and at $2x = \pi - 2\alpha$. So $\sin 2x \leq \sin 2\alpha$ exactly when $2x \leq 2\alpha$ or $2x \geq \pi - 2\alpha$. That is,

$$0 < x \leq \alpha \qquad \text{or} \qquad \tfrac{1}{2}\pi - \alpha \leq x < \tfrac{1}{2}\pi.$$

In particular, for $0 < x < \alpha$ we have $0 < 2x < 2\alpha < \frac{1}{2}\pi$, where sin is increasing, so $\sin 2x < \sin 2\alpha = k$ and $g''(x) > 0$. Hence g is convex in $0 < x < \alpha$, as required.

The other interval is

$$\tfrac{1}{2}\pi - \alpha < x < \tfrac{1}{2}\pi.$$

This is genuinely different from the first, because $\alpha < \frac{1}{4}\pi$ gives $\frac{1}{2}\pi - \alpha > \frac{1}{4}\pi > \alpha$. On the middle stretch $\alpha < x < \frac{1}{2}\pi - \alpha$, g is not convex.

Checks

- At $x = \frac{1}{6}\pi$: $\frac{2}{3}\cos^2 x = \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2} = \sin x$, so $f'' = 0$ there, as it should be at a boundary.
- In (ii), the problem is symmetric under $x \mapsto \frac{1}{2}\pi - x$, since $\sin 2x$ is unchanged. That explains why the second interval mirrors the first.
- Numerically, with $\alpha = 0.5$ ($k \approx 0.841$), a finite-difference second derivative of g is positive at $x = 0.3$ and at $x = 1.3$, and negative at $x = 0.8$. Since $\frac{1}{2}\pi - \alpha \approx 1.071$, this matches the intervals found. A computer testing many random α and x agreed with both parts.

Common slips

- Forgetting the $(h')^2$ term when differentiating $\mathrm{e}^{h(x)}$ twice. The second derivative is not just $h''\mathrm{e}^h$.
- Missing the whole of $(\pi, 2\pi)$ in (i). There $\sin x < 0 \leq \frac{2}{3}\cos^2 x$, so f is convex all the way from $\frac{5}{6}\pi$ to 2π.
- Stopping at $\sin 2x \leq \sin 2\alpha \Rightarrow x \leq \alpha$. Because sin isn't monotonic on $(0, \pi)$, this loses the second interval.

Beyond STEP

The second-derivative test is only one face of convexity. The real definition is geometric: every chord lies on or above the graph. For differentiable functions this is equivalent to f' being increasing, and it leads to Jensen's inequality, which underlies AM–GM and much of probability and optimisation. Convex functions are treated carefully in Bartle and Sherbert, *Introduction to Real Analysis*. Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

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